

CS-3180

- Welcome!
- Comparative Languages
- We will talk about a lot of them

About me

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- OSS Education
- WSU Alumni



Background:

- What languages are you all familiar with?
- A little
- A lot

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- Java
 - C++
 - Python

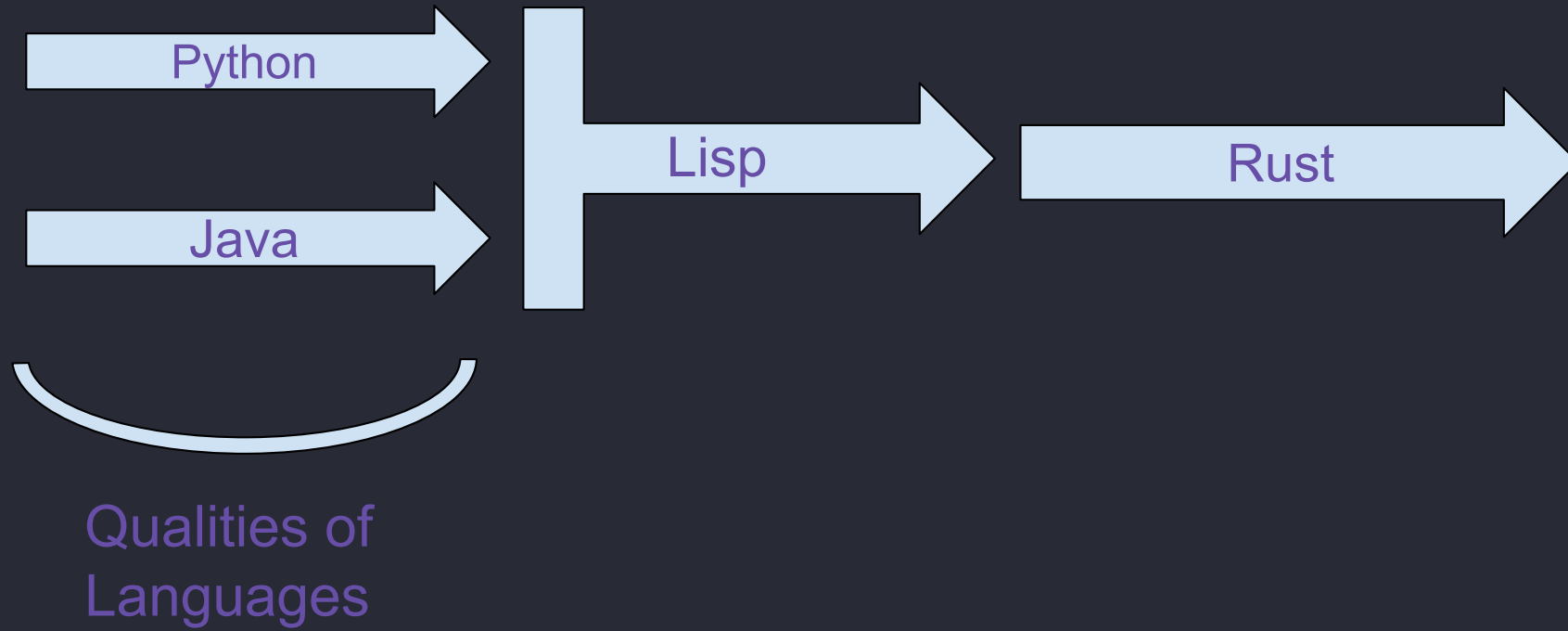
Background:

- What languages are you all familiar with?
- A little
- A lot
- Java
- C++
- Python
- HTML?

What this is about

- Similarities
- Differences
- Idiosyncrasies (isms)
- Types of languages
- Methods of execution

Timeline:



Expectations

- Syllabus
- Exams
- Detailed policies
- Available on Pilot (sorta)

What is a programming language?

- In your words?

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- Google definition:

What is a programming language?

- In your words?
- Google definition:

"A formal, structured set of instructions used to communicate with a computer and direct its behavior"

What is a programming language

- Fair enough
- Let's take a look at what fits that criteria
- Hello World's

Java Hello World

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

- A lot here (more on this later)
- Fits criteria

Python Hello World

```
a = 1 + 2 # introduce variables + printing  
print("Hello, World! ", a)
```

- Much much less (!)
- Also meets our goal
- Let's talk about this in our editor

Some Math

```
4 + (3 - 2) * 5
```

- 9

Some Math

```
4 + (3 - 2) * 5
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- Formal instruction

Some Math

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- Formal instruction
- Tell the "computer" to do something

Some Math

```
4 + (3 - 2) * 5
```

- 9
- Formal instruction
- Tell the "computer" to do something
- Directs its behavior

HTML Hello World

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-s
  <title>Hello World</title>
</head>
<body>
  <h1>Hello World</h1>
</body>
</html>
```

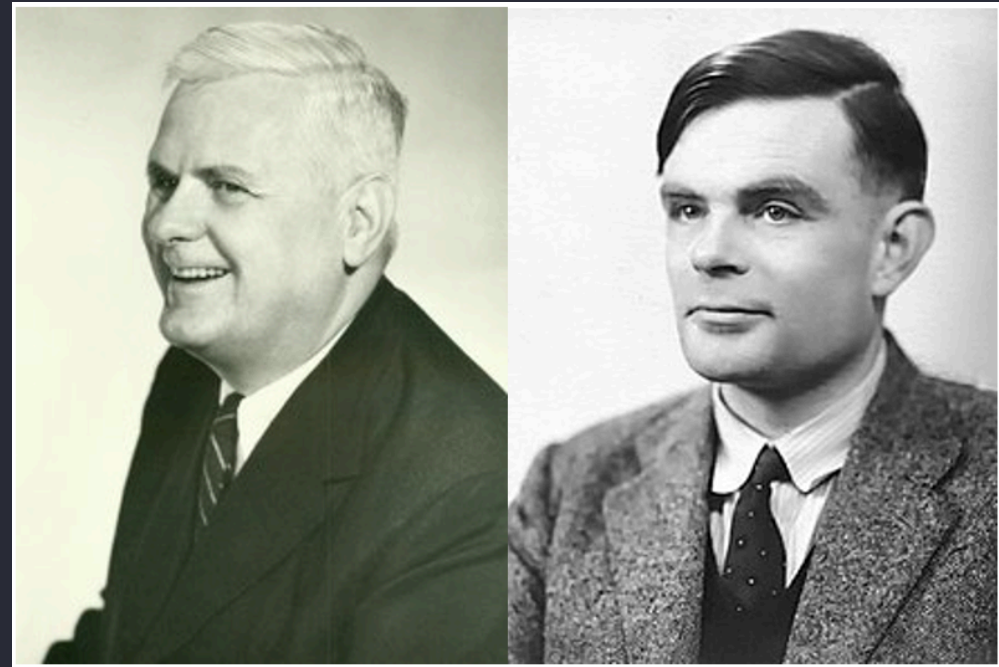
- Hyper-Text *Markup* Language
- Famously controversial

No real answer

- Vibes-based?
- Python + Java "feel" like languages
- Execute logical instructions
- HTML + Arithmetic don't
- We need a better definition
- How did we get here?

What is a computer?

- 1936
- Computer Science doesn't really exist
- Alonzo Church + Alan Turing



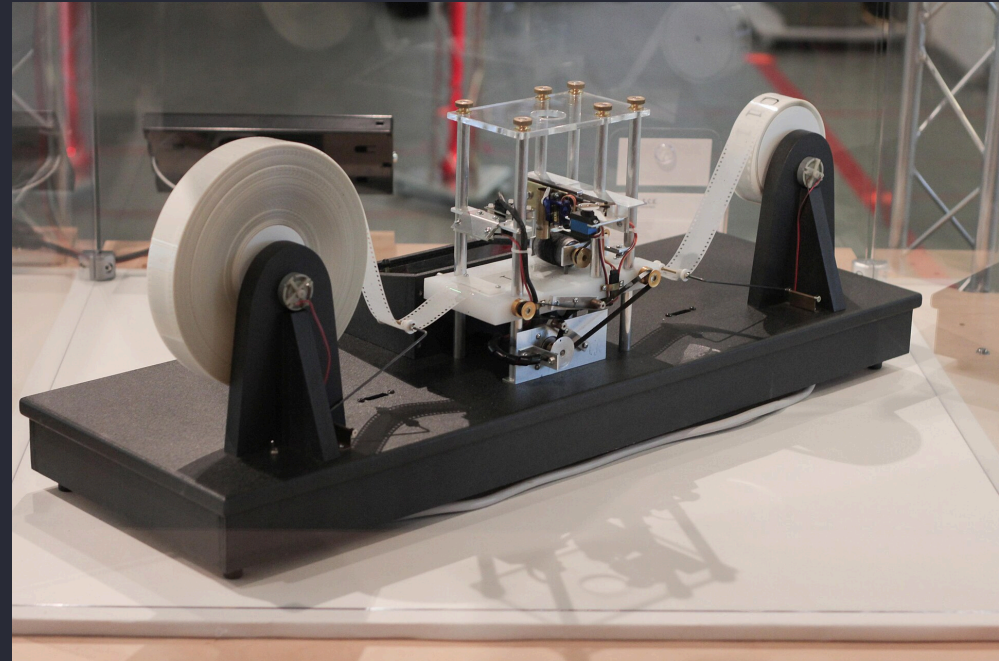
Church-Turing Thesis

- Any calculation that can be done by an effective step-by-step method can also be computed by via Turing machine
- What is a Turing Machine



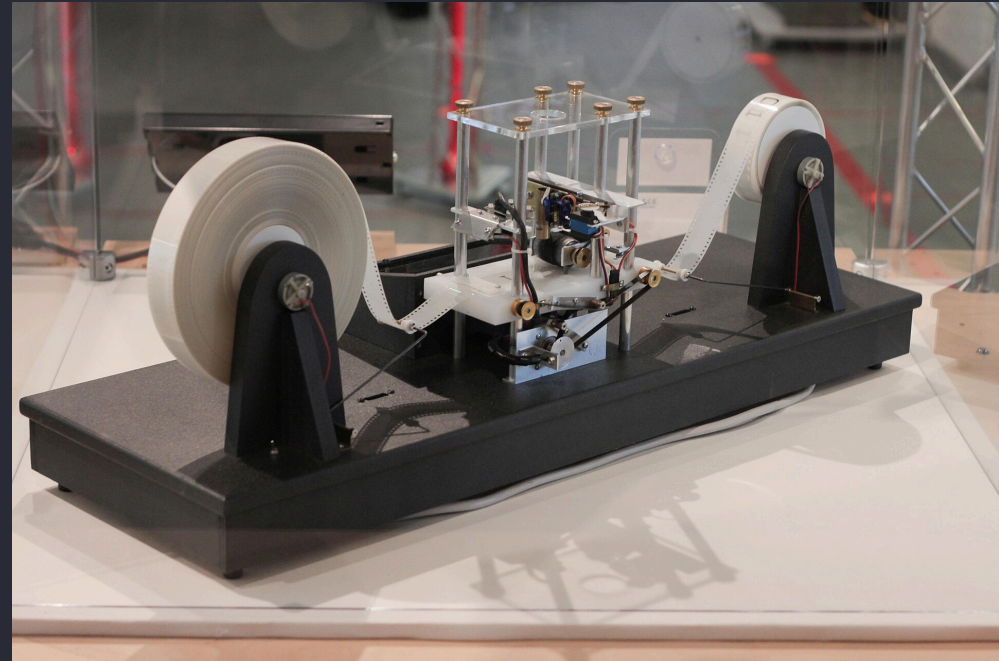
Turing Machine

- Infinite tape of cells
- Reader Head
- Interpret binary instructions
- Hypothetical



Turing Machine

- Details of this may be beyond scope
- Java + Python can simulate a turing machine
- Arithmetic + HTML cannot



Turing Completeness

- The ability to do this is referred to as "Turing Completeness"
- (more about its ability to do loops and handle state)
- Maybe this is a better definition

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- (more about its ability to do loop and handle state)
- Maybe this is a better definition
- What else fits this?

Turing Completeness

- Much more than you think
- Powerpoint
- Minecraft
- Magic the Gathering



Back to definitions

- We could add turing completeness to the definition
- Point is, it's not an easy to nail down concept
- When we talk about languages, we generally have a solid idea of what we mean, based on both definitions

Questions?